IBM FileNet P8 V5.5.8

Step-by-Step Hands-On Implementation Guide

Day 6 – Process Engine & Workflow Architecture

Process Engine Installation | PEDB Setup | Connection Point | Workflow Queues

Author	Course Day	Lab Count	Duration
Rajesh Pasham	**Day 6 of 10**	**4 Labs**	**~3.5 Hours**

LAB-06-01	LAB-06-02	LAB-06-03	LAB-06-04
Prerequisites & PEDB Setup *40 min | DB2 PEDB + JDBC Data Source*	**Install Process Engine V5.5.8** *50 min | IBM IM + PE Installation*	**Configure PE Connection Point** *35 min | ACCE Connection Point + Verify*	**Create Workflow Queues** *35 min | Roster + Work Queues + Test*

Prerequisites & Environment State

Environment State Before Day 6 Labs

All Day 6 labs build on the environment established in Days 2-5. Verify the following before starting:

CE / WAS Status	Content Platform Engine V5.5.8 running on WAS. ACCE accessible at http://localhost:9080/acce
FileNet Domain	P8Domain exists and is healthy in ACCE. ceadmin credentials work.
Object Store	TrainingOS object store exists and is accessible.
DB2 Status	DB2 V11.5 is running. GCDDB and OSDB databases exist. DB2 instance: db2admin
WAS Profile	FileNetServer profile running. Port 9080 (HTTP) and 9443 (HTTPS) active.
Free Disk Space	At least 5 GB free on C:\ drive for PE installation.
Available RAM	At least 2 GB free RAM for the PE service.
PE Software	IFCPE_5.5.8_WIN_ML.zip available in software repository (C:\Software\PE\ or network share)
IBM IM	IBM Installation Manager V1.9+ installed (from Day 2 setup)

NOTE: If any of the above prerequisites are not met, complete the relevant Day 2-5 labs before proceeding. Process Engine CANNOT be installed without a running Content Engine and valid P8Domain.

LAB-06-01 | Prerequisites & PEDB Setup

Duration: 40 Minutes | Create the Process Engine database and JDBC data source before installation

NOTE: START STATE: Content Platform Engine V5.5.8 is installed. WAS FileNetServer profile is running. DB2 V11.5 is running with GCDDB and OSDB databases existing. You are logged in as local Administrator.

Objective

In this lab you will:

- Verify all prerequisites for Process Engine installation
- Create the PEDB database in DB2 with the correct settings for FileNet PE V5.5.8
- Create the ECMProcessDS JDBC data source in WAS Admin Console
- Test the JDBC connection to verify PE can reach PEDB

STEP 1 — Verify Content Engine and WAS Are Running (5 min)

1. Open a Command Prompt as Administrator
2. Verify WAS is running by opening ACCE:

ACCE URL	`http://localhost:9080/acce`
Login	`ceadmin / ceadmin`

3. Navigate to: P8Domain > Object Stores. Verify TrainingOS is listed and accessible.
4. Verify DB2 is running:

```
db2 list db directory
# Expected output: lists GCDDB, OSDB databases

# If DB2 is not running, start it:
db2start
```

	VERIFICATION CHECKLIST
[ ]	ACCE loads at http://localhost:9080/acce without errors
[ ]	P8Domain and TrainingOS visible in ACCE
[ ]	db2 list db directory shows GCDDB and OSDB

STEP 2 — Create the PEDB Database in DB2 (10 min)

The Process Engine requires a dedicated database for storing workflow data. Unlike GCDDB and OSDB, the PEDB uses 4 KB page size.

WARNING: PEDB must use 4 KB page size (4096 bytes). This is different from GCDDB and OSDB which use 32 KB. Using 32 KB for PEDB will cause PE bootstrap to fail.

5. Open a DB2 Command Window as Administrator:
6. Start Menu > IBM DB2 > DB2 Command Window (Administrator)
7. Run the following commands exactly:

```
db2 CREATE DATABASE PEDB AUTOMATIC STORAGE YES
    USING CODESET UTF-8 TERRITORY US
    COLLATE USING SYSTEM PAGESIZE 4096

db2 CONNECT TO PEDB

db2 GRANT DBADM ON DATABASE PEDB TO USER ceadmin

db2 CREATE BUFFERPOOL PEBP4K IMMEDIATE SIZE 250 AUTOMATIC
    PAGESIZE 4096

db2 CREATE SYSTEM TEMPORARY TABLESPACE PESYSTMP PAGESIZE 4096
    MANAGED BY AUTOMATIC STORAGE BUFFERPOOL PEBP4K

db2 CREATE REGULAR TABLESPACE PEDATA PAGESIZE 4096
    MANAGED BY AUTOMATIC STORAGE BUFFERPOOL PEBP4K

db2 CREATE USER TEMPORARY TABLESPACE PETMP PAGESIZE 4096
    MANAGED BY AUTOMATIC STORAGE BUFFERPOOL PEBP4K

db2 DISCONNECT PEDB

db2 LIST DB DIRECTORY
# Verify PEDB appears in the list
```

	VERIFICATION CHECKLIST
[ ]	PEDB database created successfully (no SQL errors)
[ ]	db2 LIST DB DIRECTORY shows PEDB in the list
[ ]	PEDB page size is 4096 bytes
[ ]	ceadmin has DBADM privileges on PEDB

STEP 3 — Extract PE Software Package (5 min)

8. Locate the PE software package in your software repository:

PE Package	`IFCPE_5.5.8_WIN_ML.zip`
Location	`C:\Software\PE\ or as provided by instructor`

9. If not extracted, right-click the ZIP > Extract All to: C:\Software\PE\
10. Verify the IBM Installation Manager repository is available:

```
dir C:\Software\PE\repository.config
# This file should exist in the extracted PE package
```

	VERIFICATION CHECKLIST
[ ]	IFCPE_5.5.8_WIN_ML content extracted to C:\Software\PE\ or equivalent

[ ]	repository.config file exists in the PE software directory

STEP 4 — Create ECMProcessDS JDBC Data Source in WAS — 15 min

The Process Engine connects to PEDB through a JDBC data source named ECMProcessDS in WAS. This must be created BEFORE running the PE installer.

11. Open WAS Admin Console: https://localhost:9043/ibm/console
12. Log in as wasadmin / wasadmin
13. Navigate to: Resources > JDBC > Data Sources
14. Click New to create a new data source
15. Step 1 – Enter basic settings:

Data source name	`ECMProcessDS`
JNDI name	`ECMProcessDS`
Database type	`DB2`
Implementation type	`Connection pool data source`

16. Step 2 – Select JDBC provider: DB2 Universal JDBC Driver Provider (choose XA or non-XA)
17. Step 3 – Enter database-specific settings:

Database name	`PEDB`
Server name	`localhost`
Port number	`50000`
Driver type	`4 (Type 4 - pure Java)`

18. Step 4 – Configure authentication alias:

Component-managed authentication	`fileNetDB2Alias (create new or use existing)`
User ID	`ceadmin`
Password	`ceadmin`

19. Click Finish, then Save to master configuration
20. Click on ECMProcessDS > Test Connection

[SCREENSHOT]
WAS Admin Console showing ECMProcessDS with Test Connection result: The test connection operation was successful

TIP: If Test Connection fails with SQL4612N (auth error), verify ceadmin has DBADM on PEDB. If it fails with SQLCODE=-1024 (database not found), verify PEDB was created and DB2 is running.

VERIFICATION CHECKLIST

[ ]	ECMProcessDS data source created in WAS Admin Console
[ ]	JNDI name is exactly: ECMProcessDS (case-sensitive)
[ ]	Test Connection result: 'The test connection operation was successful'
[ ]	Data source saved to master configuration

LAB-06-02 | Install Process Engine V5.5.8

Duration: 50 Minutes | Install IBM FileNet Process Engine V5.5.8 using IBM Installation Manager

NOTE: START STATE: Lab-06-01 completed. PEDB exists in DB2. ECMProcessDS JDBC data source created and tested in WAS. WAS FileNetServer profile is running. You are logged in as local Administrator.

Objective

In this lab you will:

- Launch IBM Installation Manager and add the PE software repository
- Install IBM FileNet Process Engine V5.5.8 to C:\IBM\FileNet\ProcessEngine
- Run the PE bootstrap utility to initialize the PEDB schema
- Start the IBM FileNet Process Engine Windows service
- Verify the PE service is running and accepting connections

STEP 1 — Launch IBM Installation Manager (5 min)

1. Click Start Menu > IBM Installation Manager > IBM Installation Manager
2. OR run: C:\IBM\InstallationManager\eclipse\IBMIM.exe
3. IBM Installation Manager opens. Click File > Preferences > Repositories
4. Click Add Repository
5. Browse to: C:\Software\PE\repository.config (or the directory containing it)
6. Click OK > Apply > OK
7. Return to the main IBM IM screen

[SCREENSHOT]
IBM Installation Manager main screen with PE repository added in File > Preferences > Repositories

STEP 2 — Install Process Engine (20 min)

8. In IBM Installation Manager, click Install
9. IBM IM searches the added repository. After a moment, IBM FileNet Process Engine V5.5.8 appears in the package list
10. Check the checkbox next to IBM FileNet Process Engine 5.5.8
11. Click Next
12. Accept the license agreement > Next
13. On the Installation Directory screen, verify or set:

Installation Directory	`C:\IBM\FileNet\ProcessEngine`
Shared Resources Directory	`C:\IBM\IMShared (use existing from Day 2)`

14. Click Next to proceed to Configuration
15. Fill in the Process Engine Configuration fields:

Content Engine URI	`http://localhost:9080/wsi/FNCEWS40MTOM/`
FileNet P8 Domain	`P8Domain`
CE User Name	`ceadmin`
CE Password	`ceadmin`
PE Database Type	`DB2`
JDBC Data Source JNDI	`ECMProcessDS`
Process Engine Port	`32768 (default)`
WAS Profile Path	`C:\IBM\WAS\profiles\FileNetServer`

IMPORTANT: The Content Engine URI must be exactly: http://localhost:9080/wsi/FNCEWS40MTOM/ — this is the MTOM web services endpoint. Using the wrong URL will cause PE to fail to communicate with CE.

16. Click Next > Review the installation summary
17. Click Install to begin installation (this takes approximately 10-15 minutes)
18. Wait for the installation to complete. Progress bar shows package installation stages

[SCREENSHOT]
IBM Installation Manager showing Process Engine installation progress at approximately 75%

19. When complete, click Finish to close IBM IM

	VERIFICATION CHECKLIST
[ ]	IBM IM Installation completes with status: Process Engine installed successfully
[ ]	Directory C:\IBM\FileNet\ProcessEngine exists with PE files
[ ]	No error messages during installation

STEP 3 — Run PE Bootstrap to Initialize PEDB Schema (10 min)

The PE Bootstrap utility creates all required tables, indexes, and stored procedures in the PEDB database. This must be run once after installation.

20. Open a Command Prompt as Administrator
21. Navigate to the PE bootstrap directory:

```
cd C:\IBM\FileNet\ProcessEngine\bin
```

22. Run the bootstrap utility:

```
vwbootstrap.exe

# The bootstrap tool will:
# 1. Connect to WAS and locate ECMProcessDS
# 2. Connect to PEDB via the data source
```

```
# 3. Create all PE schema objects (tables, indexes, sequences)
# 4. Initialize default PE configuration in PEDB
# 5. Register PE with the FileNet P8 domain

# Expected output (last few lines):
# Creating Process Engine schema objects...
# Schema creation complete.
# Process Engine bootstrap completed successfully.
```

WARNING: If bootstrap fails with 'ECMProcessDS not found', verify the JNDI name in WAS is exactly ECMProcessDS (no spaces, exact case). If it fails with DB2 errors, verify PEDB exists and ceadmin has DBADM privileges.

23. Verify bootstrap created the PE tables in DB2:

```
db2 CONNECT TO PEDB
db2 LIST TABLES FOR SCHEMA PE
# Expected: Lists PE tables like VWTASKITEM, VWQUEUE, VWROSTER, etc.
db2 DISCONNECT PEDB
```

	VERIFICATION CHECKLIST
[ ]	vwbootstrap.exe completes with 'Process Engine bootstrap completed successfully'
[ ]	db2 LIST TABLES FOR SCHEMA PE shows PE tables (VWTASKITEM, VWQUEUE, etc.)
[ ]	No error messages in bootstrap output

STEP 4 — Start the Process Engine Windows Service (5 min)

24. Open Windows Services: Start Menu > services.msc
25. Scroll down to find: IBM FileNet Process Engine
26. Right-click > Start
27. Wait for the status to change to Running (approximately 15-30 seconds)

TIP: If the service fails to start, check C:\IBM\FileNet\ProcessEngine\log\vwtaskd.log for error messages. Common causes: PEDB not accessible, ECMProcessDS misconfigured, or port 32768 already in use.

28. Verify the service is listening on port 32768:

```
netstat -an | findstr 32768
# Expected output:
# TCP    0.0.0.0:32768    0.0.0.0:0    LISTENING
```

[SCREENSHOT]
Windows Services showing IBM FileNet Process Engine with Status: Running

	VERIFICATION CHECKLIST
[ ]	IBM FileNet Process Engine service status is Running in services.msc
[ ]	netstat -an | findstr 32768 shows port 32768 in LISTENING state

[ ]	No errors in C:\IBM\FileNet\ProcessEngine\log\vwtaskd.log

STEP 5 Verify PE Installation in ACCE — 10 min

29. Switch to ACCE in the browser: http://localhost:9080/acce
30. In the left navigation, expand P8Domain
31. Look for a new node: Process Engine. If not visible, click Refresh on the left tree panel.
32. Click Process Engine to expand it
33. You should see sub-nodes: Connection Points, Isolated Regions, and Rosters

[SCREENSHOT]
ACCE left navigation tree showing P8Domain > Process Engine with Connection Points, Isolated Regions, Rosters sub-nodes

	VERIFICATION CHECKLIST
[ ]	P8Domain > Process Engine node is visible in ACCE
[ ]	Sub-nodes visible: Connection Points, Isolated Regions, Rosters
[ ]	No error indicators on the Process Engine node

LAB-06-03 | Configure PE Connection Point

Duration: 35 Minutes | Create the Connection Point linking P8Domain to the Process Engine

NOTE: START STATE: Lab-06-02 completed. IBM FileNet Process Engine V5.5.8 is installed and the Windows service is Running. PEDB is initialized. ACCE shows Process Engine node under P8Domain.

Objective

In this lab you will:

- Create a PE Connection Point named PE_ConnectionPoint in ACCE
- Link the Connection Point to the TrainingOS Object Store
- Configure Isolated Region settings
- Verify the Connection Point status shows Online
- Test PE connectivity from ACCE

STEP 1 | Create a New PE Connection Point | 15 in

1. In ACCE, navigate to: P8Domain > Process Engine > Connection Points
2. Click New Connection Point in the right-hand panel
3. The New Connection Point wizard opens. Fill in the following fields:

Connection Point Name	`PE_ConnectionPoint`
Display Name	`Training Process Engine Connection`
Description	`Process Engine connection for FileNet P8 training environment`
Process Engine Server	`localhost`
Port	`32768 (default - do not change)`
Connection Type	`Standard (default)`

4. Click Next
5. On the Object Store tab, click Add
6. From the list, select TrainingOS
7. Click OK

Object Store	`TrainingOS`
Default Workflow Roster	`DefaultRoster (auto-populated)`

8. Click Next

[SCREENSHOT]
ACCE New Connection Point wizard showing Connection Point Name, Server, Port, and TrainingOS Object Store selected

STEP 2 Configure Isolated Region (5 min)

9. On the Isolated Region screen:
10. Leave the Isolated Region at the default: Region 1
11. Region 1 is the default region used for all workflow definitions in the training lab

> **NOTE:** Isolated Regions group workflow instances for performance and management isolation. For training, Region 1 is always sufficient. Production environments may create additional regions to separate HR workflows from Finance workflows, for example.

12. Click Next > Review the Summary screen
13. Verify all settings:

Connection Point Name	`PE_ConnectionPoint`
PE Server	`localhost:32768`
Object Store	`TrainingOS`
Isolated Region	`Region 1`

14. Click Finish

STEP 3 Verify Connection Point Status (10 min)

15. After clicking Finish, the Connection Point appears in the list
16. The initial status may show Connecting... — wait 10-20 seconds, then click Refresh
17. The status should change to Online (shown as a green indicator)

> **WARNING:** If the Connection Point status shows Offline or Error after 60 seconds: (1) Verify PE Windows service is running in services.msc, (2) Check firewall is not blocking port 32768, (3) Check C:\IBM\FileNet\ProcessEngine\log\vwtaskd.log for connection errors.

[SCREENSHOT]
ACCE Connection Points list showing PE_ConnectionPoint with green Online status indicator

18. Click on PE_ConnectionPoint to view its properties
19. On the Properties tab, verify:

Status	`Online`
Server	`localhost:32768`
Object Store	`TrainingOS`
Region	`Region 1`

	VERIFICATION CHECKLIST
[ ]	PE_ConnectionPoint appears in ACCE > P8Domain > Process Engine > Connection Points
[ ]	Connection Point status shows: Online (green indicator)

[ ]	Object Store TrainingOS is linked to this Connection Point
[ ]	Region 1 is assigned

STEP 4 — Test PE Connectivity via ACCE Process Configuration Console — 5 min

20. In ACCE, click Tools > Process Configuration Console in the top menu
21. The Process Configuration Console opens in a new browser tab or panel
22. If prompted to select a Connection Point, choose PE_ConnectionPoint
23. The console should load and show the PE server information:

Expected - Server Status	`Running`
Expected - Dispatch Threads	`10 (default)`
Expected - Queue Count	`1 (DefaultQueue in DefaultRoster)`

[SCREENSHOT]
ACCE Process Configuration Console showing PE_ConnectionPoint connected with Server Status Running

	VERIFICATION CHECKLIST
[ ]	ACCE Tools > Process Configuration Console loads successfully
[ ]	Console connects to PE_ConnectionPoint without errors
[ ]	Server Status shows: Running
[ ]	DefaultRoster and DefaultQueue visible in the console

LAB-06-04 | Create Workflow Queues

Duration: 35 Minutes | Create a custom Roster and Work Queues for the training workflow

NOTE: START STATE: Lab-06-03 completed. PE_ConnectionPoint is Online. Process Configuration Console is accessible via ACCE. WAS and PE services are running.

Objective

In this lab you will:

- Create a new Roster named TrainingRoster in the Process Configuration Console
- Create two Work Queues inside TrainingRoster: TrainingApprovalQueue and TrainingReviewQueue
- Configure queue participants (assign access to ceadmin and authenticated users)
- Verify queues are accessible and visible in the Process Configuration Console
- Test queue depth monitoring

STEP 1 — Open Process Configuration Console (3 min)

1. In ACCE, click Tools > Process Configuration Console
2. If PE_ConnectionPoint is not already selected, choose it from the Connection Point dropdown
3. The console opens showing the PE server configuration
4. In the left panel, you should see: Rosters (with DefaultRoster listed)

STEP 2 — Create TrainingRoster (10 min)

5. In the Process Configuration Console left panel, right-click Rosters
6. Select New Roster
7. Fill in the Roster properties:

Roster Name	`TrainingRoster`
Display Name	`Training Environment Roster`
Description	`Roster for all Day 6-10 training workflow queues`
PE Server	`localhost:32768`
Isolated Region	`Region 1 (select from dropdown)`
Maximum Workflow Instances	`0 (unlimited)`

8. Click OK to create the roster
9. TrainingRoster appears in the Rosters list in the left panel

[SCREENSHOT]
Process Configuration Console showing TrainingRoster in the Rosters list alongside DefaultRoster

	VERIFICATION CHECKLIST
[ ]	TrainingRoster appears in the Process Configuration Console Rosters list

STEP 3 — Create TrainingApprovalQueue (10 min)

10. In the left panel, expand TrainingRoster
11. Right-click TrainingRoster > New Queue
12. Fill in the Queue properties for the Approval Queue:

Queue Name	`TrainingApprovalQueue`
Display Name	`Training Approval Queue`
Description	`Documents awaiting management approval`
Queue Type	`Classic (FIFO - default)`
Maximum Depth	`100 (alert threshold)`

13. On the Participants tab, click Add Participant
14. Search for user: ceadmin. Add it as a participant with Full Access.
15. Also add #AUTHENTICATED-USERS as participants:

Participant 1	`ceadmin — Full Access (can view, claim, step, reassign)`
Participant 2	`#AUTHENTICATED-USERS — View + Step (can view and step but not reassign)`

16. Click OK to create the queue

[SCREENSHOT]
Process Configuration Console showing TrainingApprovalQueue created under TrainingRoster with ceadmin as participant

STEP 4 — Create TrainingReviewQueue (8 min)

17. Right-click TrainingRoster > New Queue again
18. Fill in the Queue properties for the Review Queue:

Queue Name	`TrainingReviewQueue`
Display Name	`Training Review Queue`
Description	`Documents awaiting technical review before approval`
Queue Type	`Classic (default)`
Maximum Depth	`50`

19. Add participants on the Participants tab:

Participant	`ceadmin — Full Access`

Participant	`#AUTHENTICATED-USERS — View + Step`

20. Click OK to create the queue

	VERIFICATION CHECKLIST
[ ]	TrainingApprovalQueue created under TrainingRoster
[ ]	TrainingReviewQueue created under TrainingRoster
[ ]	Both queues visible in the Process Configuration Console
[ ]	ceadmin listed as participant in both queues

STEP 5 — Verify Queue Structure in ACCE (4 min)

21. In ACCE, navigate to: P8Domain > Process Engine > Rosters
22. Verify TrainingRoster is listed alongside DefaultRoster
23. Expand TrainingRoster — verify TrainingApprovalQueue and TrainingReviewQueue are listed
24. Check the Queue Depth for each queue (should be 0 — no work items yet):

TrainingApprovalQueue	`Current Depth: 0 (expected — no workflows launched yet)`
TrainingReviewQueue	`Current Depth: 0 (expected — no workflows launched yet)`

[SCREENSHOT]
ACCE showing P8Domain > Process Engine > Rosters > TrainingRoster with both queues listed

STEP 6 — Test Queue Depth Monitoring (0 min)

The Process Configuration Console shows live queue depth and allows you to monitor workflow loads. In this step, explore the monitoring capabilities:

25. In Process Configuration Console, click on TrainingApprovalQueue
26. The right panel shows queue properties and statistics:

Current Depth	`0 (no work items queued)`
Maximum Depth	`100 (alert threshold set in Step 3)`
Queue Type	`Classic`
Participants	`ceadmin, #AUTHENTICATED-USERS`

27. Note the Refresh button — in production, click Refresh to see real-time queue depth updates
28. This view will show active work items once workflows are created and launched in Day 7

> **TIP:** In production monitoring, set up a dashboard that refreshes the Process Configuration Console every 5 minutes. If any queue depth exceeds 50% of its Maximum Depth threshold, investigate the cause — either users are not processing items or a component step is failing.

Day 6 Lab Completion Summary

Process Engine is now installed, configured, and ready for workflow development in Day 7.

Lab	Title	Skills Practiced	Status
LAB-06-01	Prerequisites & PEDB Setup	DB2 PEDB, JDBC ECMProcessDS	**[ ] Complete**
LAB-06-02	Install Process Engine V5.5.8	IBM IM, bootstrap, PE service	**[ ] Complete**
LAB-06-03	Configure PE Connection Point	ACCE Connection Point, Online status	**[ ] Complete**
LAB-06-04	Create Workflow Queues	Roster, Work Queues, Participants	**[ ] Complete**